



UNIVERSITY OF PERADENIYA
DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE
END SEMESTER EXAMINATION, SEMESTER II – October, 2016
ST102 – INTRODUCTION TO PROBABILITY THEORY – PART I.

Time Allowed: **TWO Hours**

Answer **All Questions.**

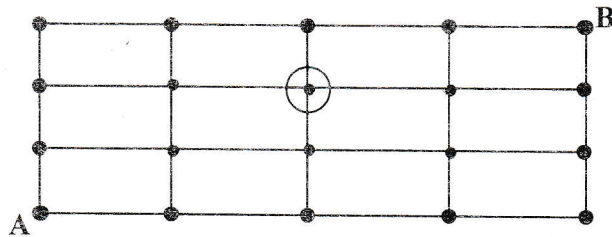
1.

a). Show that

$$k \binom{n}{k} = (n - k + 1) \binom{n}{k-1} = n \binom{n-1}{k-1}$$

b). Consider a set of n antennas of which m are defective. Suppose that all of the defectives and all of the non defective antennas are considered indistinguishable. How many linear orderings are there in which no two defectives are consecutive?

c). Consider the grid of points shown below. Suppose that starting from point “A”, you can go one step up or one step to the right at each move. This is continued until the point “B” is reached.



What is the probability that the chosen path contains the point circled in the figure?

2. Let X be a continuous random variable with the following probability density function (pdf).

$$f(x) = \begin{cases} \frac{x}{25} & ; 0 \leq x < 5 \\ \frac{2}{5} - \frac{x}{25} & ; 5 \leq x < 10 \\ 0 & ; \text{o/w} \end{cases}$$

a). Show that $f(x)$ is a legitimate pdf.

b). Find $E(X)$ and $Var(X)$.

c). Derive the Cumulative Density Function (CDF) of X .

d). Calculate $P(3 < x < 6)$.

e). Calculate $P(5 < x < 8 | x > 7)$.

[P.T.O.]

- 3.
- Suppose X has a binomial (n, p) distribution and let Y have a negative binomial (r, p) distribution. Show that $P[x < (r - 1)] = 1 - P[y < (n - r)]$.
 - Two movie theaters compete for the business of 1000 customers. Assume that each customer chooses between the movie theaters independently and with equally likely. Let N denote the number of seats in each theater.
 - Using a binomial model, **find an expression** for N that will guarantee that the probability of turning away a customer (because all seats are full) is less than 1%.
 - Use the normal approximation to get a numerical value for N .
4. Let a continuous random variable X follows uniform distribution within the interval $[\alpha, \beta]$.

$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & ; \alpha \leq x \leq \beta \\ 0 & ; \text{o/w} \end{cases}$$

- Derive the Moment Generating Function (MGF) of X .
- Derive an expression for the k^{th} moment about the origin.
- Show that the mean and median are same for X .
- You arrived at a bus stop at 10'o clock, knowing that the bus will arrive at some time uniformly distributed between 10 and 10:30. If at 10:10 the bus has not yet arrived, what is the probability that you will have to wait at least an additional 15 minutes?

Distribution	Density Function
Binomial (n, θ)	$P(X = x) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}; x = 0, 1, 2, \dots, n$
Negative Binomial (k, θ)	$P(X = x) = \binom{x-1}{k-1} \theta^k (1 - \theta)^{x-k}; x = k, k+1, k+2, \dots$
Geometric (θ)	$P(X = x) = \theta^x (1 - \theta)^{x-1}; x = 1, 2, \dots$